

SAM TSANG

Medical Robotics & AI Engineer — MSc, University College London

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PROFILE

Engineer who builds real medical devices and proves they work with measurements — an MR-safe pneumatic actuator for robot-assisted endovascular surgery, an EMG-driven soft exoskeleton, and a vision-controlled prosthetic hand. Strong across mechatronic prototyping (pneumatics, Arduino firmware, drive electronics, 3D printing) and software (Python, C++, MATLAB, R, PyTorch, OpenCV, ROS 2).

EDUCATION

MSc Medical Robotics & Artificial Intelligence — University College London Sep 2025 – Sep 2026
Robotic Systems Engineering · Medical Robotics & Devices · AI for Surgery · Surgical Data Science · ML in Medical Imaging · Robot Vision & Navigation

BEng Biomedical Engineering — University of Reading — Upper Second Class Honours (2:1) Sep 2022 – Jun 2025

EXPERIENCE

Research Student — MR-Safe Surgical Robotics UCL Medical Physics & Biomedical Engineering Oct 2025 – Jul 2026

- Engineered the complete pneumatic and electrical drive system for a three-chamber rolling-diaphragm rotary actuator for MR-guided catheter robots: solenoid-valve manifold, MOSFET switching electronics, and Arduino firmware executing 120°-phased valve sequencing at a 60 ms period.
- Characterised the full operating envelope on hardware — 0.18–0.60 N·m torque, 169–215 rpm, 73–83% efficiency, 12.4–43.0 N thrust, 190 s continuous rotation — exceeding the clinical catheter requirement by over 10×.
- Traced the dominant pneumatic loss to the solenoid valve (27% pressure drop) through systematic circuit mapping, redirecting the redesign priority; delivered an MR-safe build with all in-bore parts non-metallic and 3D-printed (ASTM F2503).

Summer Research Intern — Biomedical Data Science CUHK-Shenzhen, Futian Biomedical Innovation R&D Centre Jul – Aug 2025

- Built automated R pipelines to clean, deduplicate, standardise and merge multi-source clinical datasets covering 80,000+ patient records, parallelised with R's *parallel* package.
- Automated subgroup statistical analysis and publication-quality figure generation (chi-square, multinomial logistic regression, non-parametric tests, ggplot2), cutting repeat analyses from hours to minutes; developed Python web-crawling scripts for research-data collection.

SELECTED PROJECTS

Surgical Workflow Recognition — Cholec80 (103 GB). PyTorch ResNet-18 phase recognition with leakage-safe video-level splits; diagnosed majority-class mode collapse, applied weighted loss, built a timed-fusion model (training loss 1.77 → 0.06), validated with paired Wilcoxon tests. PyTorch · CNNs · Python

Visual SLAM — implementation & ablation. EKF and factor-graph SLAM in MATLAB; systematic ORB-SLAM2 (C++) ablations on KITTI-07 and TUM RGB-D, patching Tracking.cc; Sim(3)-aligned ATE of 2.84 m outdoor / 5.1 cm indoor. C++ · SLAM · MATLAB

Manipulator kinematics, dynamics & planning. Checkpoint-following pipeline for a 5-DOF arm in Python/ROS 2 — rosbag2 extraction, KDL forward kinematics, SLERP interpolation, damped-least-squares IK — plus 7-DOF dynamics derived from first principles. ROS 2 · Python · Kinematics

Webcam-controlled robotic prosthetic hand. Real-time gesture replication on a 3D-printed Brunel Hand 2.0 with five Dynamixel AX-12A motors; OpenCV + MediaPipe tracking mapped to motor commands. *Department showcase selection.* OpenCV · MediaPipe · Dynamixel

EMG-controlled soft exoskeleton. Wearable forearm assistance using McKibben pneumatic muscles triggered by three MyoWare EMG sensors via Arduino and a MOSFET-switched pump; four actuator generations validated for force, range of motion and wearability. EMG · Pneumatics · Arduino

TECHNICAL SKILLS

Programming	Python · C++ · MATLAB · R · SQL
Robotics & Control	ROS 2 · SLAM · kinematics & dynamics · trajectory planning · pneumatics · Arduino / embedded · EMG interfacing · real-time control
ML & Vision	PyTorch · OpenCV · MediaPipe · scikit-learn · CNNs · statistical analysis
Hardware & Tools	3D printing · Fusion 360 · drive electronics · soldering · Dynamixel · Raspberry Pi · Git · Linux

ADDITIONAL

Languages: English, Cantonese, Mandarin — all native · **Awards:** Silver Medal, UK Mathematics Olympiad · Finalist, University of Reading Project Showcase · **Certificates:** Introduction to Robotics (Coursera) · Python for Data Science (edX)